

Building StockTrader: Signals, Strategies, and Disagreement

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https://github.com/afro-chai/AGAI_HW2-IndividualAssignment-Coding

Abstract

This work implements a small multi-agent **StockTrader** pipeline: two graded strategies—**News Sentiment Follower** and **Volatility Averse**—plus a third **Moral Trader** extension, orchestrated with Microsoft **AutoGen** and local **Ollama** inference. Each symbol is analyzed in parallel from the same market snapshot; an evaluator then summarizes agreement and tension across agents. The study uses a compact basket of five U.S. listings chosen for sector spread and defense–geopolitics exposure, including Africa–MENA energy and gold mining angles alongside defense and shipping names. A lightweight news sentiment feed supplies optional headlines; a separate historical exercise scores transparent stand-ins for the two graded philosophies against realized forward returns without replaying the LLMs on past dates.

1 Strategy Selection and Rationale

The two graded strategies I implemented are **News Sentiment Follower** (Strategy A) and **Volatility Averse** (Strategy B). I paired them because they differ on **what counts as evidence**: Strategy A privileges external narrative and sentiment when available, while Strategy B treats realized turbulence, drawdown depth, and stability of the tape as first-class risk signals. I chose them because of the inherent applicability to the defense sector.

Defense–geopolitics framing. The default run uses a **small, diverse** basket with explicit **AFRICOM** hooks on **TTE** (Africa–MENA oil and LNG) and **GOLD** (Mali / DRC gold exposure narratives), plus U.S. defense software, a prime, and tanker beta.

Extension. Moral Trader adds a third parallel branch for future accountability, evaluated only after A and B finish for the same ticker in hopes of future final project incorporation.

Expected disagreement (before running). I expected the sharpest splits where *(i)* headline sentiment is missing or stale but price paths are extreme, because News Sentiment must lean on sparse momentum proxies while Volatility Averse still reads realized turbulence and drawdowns; *(ii)* momentum looks strong yet short-horizon volatility looks “calm,” tempting a BUY from narrative-style reasoning vs. a HOLD from a risk gate; and *(iii)* Moral Trader sometimes amplifies API or data-integrity notes, producing a third axis that does not track A/B mechanically. Defense, energy, mining, and tankers were chosen partly to force those regimes to appear in one batch.

2 System Architecture

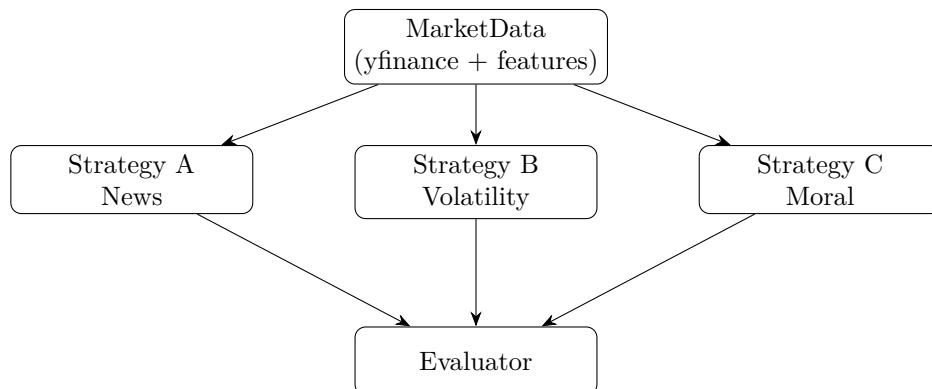
Flow (plain language). For each ticker the pipeline builds one structured market payload (prices, volatility, optional news summaries), fans that payload out to three strategy branches

that run concurrently, then passes all three structured outputs into a final evaluator agent. No strategy sees another strategy’s text during the decision pass; only the evaluator compares them. Artifacts are written per ticker plus a run-level summary and an optional historical scorecard.

2.1 Orchestration logic

- **Market data (no LLM):** Python services pull multi-window OHLC-derived features from yfinance and optionally attach Alpha Vantage `NEWS_SENTIMENT` (one request per symbol per run when configured).
- **Strategy agents (LLM):** three AutoGen `AssistantAgent` instances per ticker, each with its own system prompt, run in parallel and return structured JSON (decision, confidence, justification).
- **Isolation:** all three receive the same market JSON; they do not exchange messages before evaluation.
- **Evaluator (LLM):** consumes the three structured outputs and emits agreement flags plus narrative comparison.
- **Persistence:** one JSON bundle per ticker, a cross-ticker summary file, and an optional deterministic backtest file for the assignment bonus.

2.2 Diagram



A more comprehensive architecture breakdown can be found by downloading the html files and opening the `Stock_Trader_Walkthrough_Home.html` in the walkthrough folder located at `report/walkthrough` in the repository.

3 Stock Selection and Rationale

I chose the following stocks listed in the table below to analyze because of their relevance to the defense sector along with the diversity in their respective volatility as compared to one another.

Ticker	Company (short)	Sector / theme	Vol. guide
PLTR	Palantir	AI / gov analytics	High
TTE	TotalEnergies	Oil / LNG; AFRICOM Africa–MENA	Low
GOLD	Barrick Gold	Mining; AFRICOM Mali / DRC gold	Medium
LMT	Lockheed Martin	Defense prime	Low
FRO	Frontline	Crude tankers / Hormuz beta	High

Table 1: Default stock selection with defense relevance, explanation, and volatility scores.

Why each name maps to a different condition. **PLTR** stresses high-beta growth / gov-tech where tape can be soft while headline flow is erratic. **TTE** is a macro-energy case: strong 30-day momentum can coexist with low realized volatility, inviting narrative vs. risk-gate tension. **GOLD** is a commodity drawdown + elevated risk read (sharp negative month, higher beta) where both graded agents might still align on risk-off. **LMT** is a quality-defense “moderate tape” control: mild drawdown and calmer turbulence statistics. **FRO** is shipping / freight beta: modest spot return but elevated path risk, a natural stress test for whether “calm” is defined the same way in news-leaning vs. vol-leaning prompts.

4 Results by Stock

Quantitative overview. Table 2 lists Strategy A (News Sentiment Follower), Strategy B (Volatility Averse), and Strategy C (Moral Trader) for the final Ollama run on five tickers (spread = max – min confidence across A/B/C). Each subsection gives both **graded** agents’ decisions and confidences, at least two verbatim-style justification pulls from the saved JSON, and the evaluator’s written synthesis for that symbol.

Ticker	A (News)	Conf.	B (Vol.)	Conf.	C (Moral)	Conf.	Spread
PLTR	HOLD	6	SELL	8	SELL	10	4
TTE	BUY	8	SELL	8	HOLD	3	5
GOLD	SELL	8	SELL	8	SELL	5	3
LMT	HOLD	6	SELL	8	SELL	1	7
FRO	BUY	8	SELL	8	HOLD	3	5

Table 2: Strategy A/B/C decisions and confidences

PLTR

News Sentiment: HOLD (6). **Volatility Averse:** SELL (8). **Moral Trader:** SELL (10).

“No usable news available... Current price is down by -15.53% from the previous 30-day period, but volatility is low at 0.034412... HOLDing the current position is the most conservative approach.”

“We would reduce exposure: the path has been rough—large worst-day loss and a double-digit one-month drawdown with choppy daily moves... We prefer SELL in such conditions...”

Evaluator (excerpt): “The News Sentiment Follower focuses on the absence of usable news... the Volatility Averse agent emphasizes elevated turbulence signals... stricter than the News Sentiment Follower’s implicit threshold...” (**pattern_note:** Vol & Moral SELL; News HOLD).

TTE

News Sentiment: BUY (8). **Volatility Averse:** SELL (8). **Moral Trader:** HOLD (3).

“High price momentum (15.17% in 30 days) and positive sentiment score (0.05)... news features are neutral... we lean towards buying TTE stock.” (Sentiment score is not present in JSON when **articles_count** = 0.)

“The path has been rough with a double-digit one-month drawdown (15.17% change) and choppy daily moves... We prefer SELL in this scenario to maintain defensive positioning.”

Evaluator (excerpt): *“The former prioritizes high price momentum... positive sentiment score over turbulence thresholds, whereas the latter focuses on elevated volatility... potential overbought conditions.”* (**pattern_note**: News BUY · Vol SELL · Moral HOLD).

GOLD

All three agents: SELL (News 8, Volatility 8, Moral 5).

“No usable news available. pct_change_30d is -12.86... volatility_30d is 0.032859, which is moderate... we lean towards selling...”

“We would reduce exposure: the path has been rough—large worst-day loss and a double-digit one-month drawdown... high volatility (0.032859)...”

Evaluator: *“The three strategies converged on selling...”* (**pattern_note**: Unanimous SELL (News, Vol, Moral))—the **only** three-way agreement in this run.

5 Patterns of Agreement and Disagreement

Where they agreed. Three-way unanimity appears **only on GOLD** (all SELL, confidences 8/8/5). Both graded agents and Moral read drawdown plus turbulence as defensive exit; this is the single line in `summary.json` with **agree**: `true` for all three.

Where they disagreed. **PLTR:** News HOLD vs. Volatility and Moral SELL—Moral assigns maximum confidence (10) while flagging empty news and macro beta. **TTE:** full triangle BUY / SELL / HOLD: News chases momentum language (and again cites a phantom sentiment score), Volatility now **SELLs** on “rough path” / overbought framing, Moral HOLD on empty articles. **LMT:** News HOLD vs. Volatility and Moral SELL on turbulence narrative despite moderate -5.5% month. **FRO:** same BUY / SELL / HOLD pattern as TTE with momentum vs. defensive vol language.

My read (what drives divergence). The dominant driver remains **empty news** plus **inconsistent fallback rules**: Strategy A still fabricates or stretches sentiment when fields are null (TTE, FRO), while Strategy B often converges on SELL using vol-path templates that sometimes **contradict** the sign of `pct_change_30d` (FRO). Moral Trader tracks governance notes and beta language but can emit confidence extremes (PLTR 10, LMT 1) and occasional field errors.

6 Failure or Surprise Case

Primary failure (unexpected output): **TTE** (and **FRO**) still cite positive “sentiment” or narrative where `avg_sentiment_score` is null and `articles_count` = 0. That remains **fabricated grounding** relative to the payload.

Secondary weakness: **LMT** evaluator **analysis** mis-describes Moral as HOLD though JSON says SELL; Moral text also mis-states β vs. the numeric `beta_proxy_spy`. Hard schema checks or post-validation would catch this before submission narratives.

Positive surprise: **GOLD** shows clean three-way SELL alignment with coherent risk-off language across agents—the one case where empty news did not prevent convergent de-risk framing.

7 Reflection

Which graded strategy for \$10,000 over one month? I would follow **Volatility Averse (Strategy B)**. This run showed Strategy A can drift from supplied fields when news is empty (TTE), while Strategy B stayed anchored to explicit turbulence and drawdown checks on

PLTR/FRO. For capital preservation over a short horizon, I prefer a rule-like risk gate I can audit numerically, even when it is conservative.

A third hybrid (best of both). Same as prior revision: (1) blend only when articles and scores exist; (2) if news is empty, forbid free-form sentiment; (3) validate evaluator text against structured decisions (would fix the LMT mismatch). This run proves why: four of five tickers are multi-way splits under identical feeds.

Bonus artifact: backtest snapshot

outputs/backtest.json (five tickers; **115** row samples in rows) lists heuristic hit rates with overall_winner = news_sentiment_heuristic, overall_hit_rate_news = 0.4348, overall_hit_rate_volatility = 0.2522.

Ticker	News hit	Vol. hit	Heuristic winner
PLTR	0.2174	0.4348	volatility_averse_heuristic
TTE	0.5652	0.1304	news_sentiment_heuristic
GOLD	0.5217	0.2609	news_sentiment_heuristic
LMT	0.3913	0.2174	news_sentiment_heuristic
FRO	0.4783	0.2174	news_sentiment_heuristic

Table 3: From outputs/backtest.json per_ticker

Contrast with live agents. The backtest scores transparent heuristics on historical rows; live agents consume today’s summarized JSON—the objects differ by design (see the **disclaimer** field in backtest.json).